

EXECUTIVE MEMO

Portfolio Health, Underwriting Segmentation, and Vintage Performance

To: Chief Risk Officer; Head of Underwriting
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Subject: Portfolio Health, Underwriting Segmentation, and Vintage Performance

1. SUMMARY

Analysis of 8,500 loan applications and 165,778 monthly performance snapshots on the synthetic lending portfolio. Three questions were addressed:

- What is the current state of the loan book?
- Which underwriting attributes show the clearest observed association with default?
- Are newer loan cohorts performing worse, or are they simply less seasoned?

Key findings:

- 51.84% of the \$71.36M outstanding balance sits at 90+ days past due. Over half the book is in default-tier status by the project's operational definition.
- Credit score shows the clearest observed association with 90+ DPD in the analyzed segments. DTI shows a weaker and less consistent association, while income and loan term show more limited differentiation in the analyzed cuts.
- Pricing rises with observed credit risk, but pricing adequacy cannot be established from this dataset — the required inputs (loss given default, recovery rates, funding cost, capital cost, target margin) are out of scope.
- At common MOB checkpoints, older vintages show modestly higher cumulative 90+ DPD rates than newer vintages after accounting for cohort maturity.

2. PORTFOLIO HEALTH

Metric	Value
Applications	8,500
Requested capital	\$190.29M
Approved capital	\$142.42M (74.84% conversion)
Outstanding balance	\$71.36M
90+ DPD exposure	\$36.99M (51.84% of outstanding)

60-month loans carry 58.95% of total 90+ DPD exposure, despite being the smaller share of originations. This is an exposure distribution observation — it does not establish that term causes default.

3. UNDERWRITING & RISK SEGMENTATION

Credit score — clearest observed association

90+ DPD loan rate falls monotonically from 44.2% (Poor, <600) to 28.7% (Excellent, ≥720). The same directional pattern is visible within the DTI bands in the credit-score × DTI cross-tabulation.

DTI — weaker, less consistent association

Observed 90+ DPD rises through the first three DTI bands (<20%, 20–35%, 35–50%), then drops in the ≥50% band. The ≥50% band contains only 154 loans, so the lower observed rate should be interpreted cautiously and should not be treated as evidence of a genuine risk inversion.

Income and loan term — limited segment-level differentiation

Lower income groups show higher observed 90+ DPD rates, declining smoothly from 40–43% (under \$50K) to 28–30% (\$120K+). Within every income band, 60-month terms show 1–2 percentage points higher observed rates than 36-month terms. Neither variable shows strong additional differentiation in the segment-level analysis presented here.

Pricing — descriptive only

Average interest rate falls from 16.55% (Poor) to 11.14% (Excellent), moving in the same direction as observed risk. The data does not contain expected loss, funding cost, or capital cost, so no conclusion about pricing adequacy can be drawn.

4. VINTAGE / COHORT PERFORMANCE

Measured against the full originated cohort at common Months-On-Book (MOB) checkpoints (6, 12, 18, 24), older vintages show modestly higher cumulative 90+ DPD rates than newer vintages at MOB24:

Vintage	Cumulative 90+ DPD at MOB24
2022	41.78%
2023	39.90%
2024	37.87%

The 2025 cohort is not yet mature enough for MOB18 and MOB24 comparison and is therefore excluded from those checkpoints.

An earlier version of this analysis showed the opposite pattern — later-MOB default rates appearing to decline. That result was an artifact of survivorship bias: because loan performance snapshots stop at first default, defaulted loans were dropping out of the denominator at later checkpoints. The corrected analysis uses the full cohort as the denominator and counts each loan as defaulted from its first 90+ DPD month onward.

Conclusion: the analysis provides no evidence of underwriting deterioration after controlling for cohort maturity.

5. ANALYTICAL IMPLICATIONS

- Credit score warrants continued emphasis in underwriting review, given its consistent segment-level association with observed 90+ DPD rates in this analysis.
- DTI remains useful as a secondary attribute in the 20–50% range. The ≥50% band should not drive policy changes based on this dataset — the segment is too thin.

- Income thresholds and term caps show limited differentiation in the segments analyzed and may not justify further policy complexity without additional evidence.
- To evaluate pricing adequacy, the analysis would need to be extended with: loss given default, recovery assumptions, funding cost, operating cost, capital cost, and target margin.

6. LIMITATIONS

- Synthetic data. Patterns here illustrate methodology and are not evidence about any real lending portfolio.
- 90+ DPD as operational default. This is the project's defined threshold, not a universal industry standard.
- Segment sizes vary. Conclusions about the $\geq 50\%$ DTI band and several sub-segments rest on fewer than 100 loans. Findings from thin segments are directional only.
- Associations, not causes. This analysis identifies which attributes correlate with observed 90+ DPD in the analyzed cuts. It does not establish that any attribute causes default, nor does it evaluate incremental predictive power across attributes.